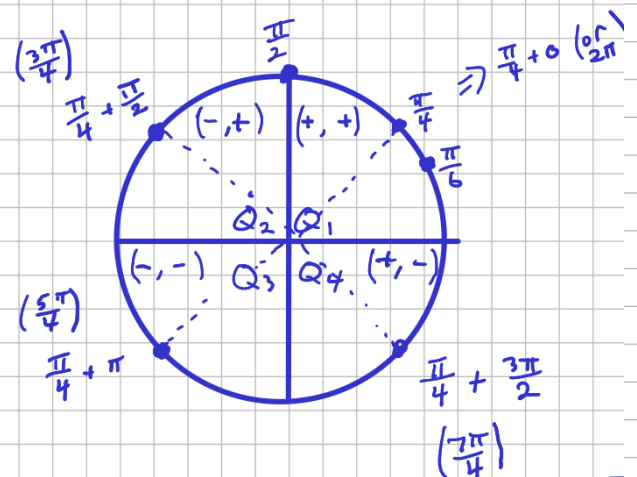


Concept	Strategy
1. Complex Numbers	1.
a) Multiplication by j	a) Multiplication of a number by j shifts its powers cyclicly. $j^2 = -1$, $j^3 = -j$, $j^4 = 1$ Geometrically, this is a 90° ccw rotation in the complex plane.
b) Division $\frac{z_1}{z_2}$	b) Multiply the numerator and denominator by the complex conjugate of z_2 . Example: $\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)}$
c) Form conversion Rectangular: $a + jb$ Polar: $r(\cos \theta + j \sin \theta)$ Exponential: $re^{j\theta}$	c) Modulus $r = z = \sqrt{a^2 + b^2}$ Principal arg θ : - Get reference angle using known parameters. Ex: $\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{\text{Im}}{r}$ $\Rightarrow$ Check the actual quadrant and adjust the angle.



d) A phasor is a snapshot of a sinusoidal signal in time - it takes a time domain signal and converts it to the frequency domain

$$x(t) = A \cos(\omega t + \theta)$$

$$X = A e^{j\theta} = A \angle \theta$$

Sinusoid:

$$f(x) = A \sin(B(x - \alpha)) + C$$

Amplitude $\uparrow$ Period $= \frac{2\pi}{|B|}$ Vertical offset/shift $\uparrow$
 Phase shift $\uparrow$ α
 $x - \alpha \rightarrow$ shifted right
 $x + \alpha \rightarrow$ shifted left

e) Conversion

e) Conversion

$$x(t) = \operatorname{Re}(X e^{j\omega t})$$

Example:

$$\begin{aligned} \operatorname{Re}(X e^{j\omega t}) &= \operatorname{Re}(A e^{j\theta} \cdot e^{j\omega t}) \\ &= \operatorname{Re}(A e^{j(\omega t + \theta)}) \end{aligned}$$

$$\Rightarrow \text{Remember } e^a \cdot e^b = e^{(a+b)}$$

$$= \operatorname{Re}(A [\cos(\omega t + \theta) + j \sin(\omega t + \theta)])$$

$$= A \cos(\omega t + \theta)$$

f) Calculations of Powers z^n

f) De Moivre's Formula

$$z^n = r^n (\cos(n\theta) + j \sin(n\theta))$$

$$\text{Ex: } z^6 = (1 + j)^6$$

$$\Rightarrow r = |z| = \sqrt{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

$$z^6 = r^6 (\cos(6\theta) + j \sin(6\theta))$$

$$= \sqrt{2}^6 (\cos(6\theta) + j \sin(6\theta))$$

$$= 8 (\cos(6\frac{\pi}{4}) + j \sin(6\frac{\pi}{4}))$$

$$= 8 (\cos(3\pi/2) + j \sin(3\pi/2))$$

$$\Rightarrow \text{ref angle: } \frac{\pi}{2}, \text{ Q3}$$

$$8(0 - j) = -j8$$

2. Differential Equations

a) 1st-order homogeneous
(unforced - decay models)

$$y'(t) + ay(t) = 0$$

b) 1st-order Forced

$$y'(t) + ay(t) = b$$

* The Differential equations
must be in the form
shown here!!!
....

c) 2nd-order (Damping class.)

$$y''(t) + ay'(t) + by(t) = 0$$

a)

1. Identify a and initial value $y(0)$
2. Solution: $y(0)e^{-at}$
3. $\tau = 1/a$
4. System is at final state when $t \approx 5\tau$

b)

1. Homogeneous solution:
 $y_h(t) = Ce^{-at}$
2. Particular solution:
 $y_p(t) = \frac{b}{a}$
3. Total solution:
 $y(t) = Ce^{-at} + \frac{b}{a}$
4. Initial condition:
 $y(0) = C + \frac{b}{a}$
 $C = y(0) - \frac{b}{a}$

$$5. y(t) = \left(y(0) - \frac{b}{a}\right)e^{-at} + \frac{b}{a}$$

c)

1. Use the characteristic equation
 $s^2 + as + b = 0$
2. Find the roots s_1 & s_2

3. 3 possible outcomes:

$$\underline{s_1 \neq s_2}$$

$$\text{Soln: } y(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

Damping: Overdamped

$$\underline{s_1 = s_2}$$

$$\text{Soln: } y(t) = (C_1 + C_2 t) e^{s_1 t}$$

Damping: Critically damped

$$\underline{s = \sigma \pm j\omega}$$

$$\text{Soln: } y(t) = e^{\sigma t} [C_1 \cos(\omega t) + C_2 \sin(\omega t)]$$

Damping: underdamped

3. Laplace Transforms

Laplace transform simply takes a calculus problem in the time domain and converts it into an algebra problem in the frequency domain.

$$1. \mathcal{L}\{k\} = \frac{k}{s}$$

$$2. \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$3. \mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$4. \mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$$

$$5. \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

Properties:

1. Linearity

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$$

2. Shift

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

3. Derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$